

SANJAI K

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PROFESSIONAL SUMMARY

Passionate about transforming data into meaningful insights, I am an MCA graduate with IEEE-published research and hands-on experience across Machine Learning, Deep Learning, Data Engineering, and Data Analytics. Skilled in building end-to-end ML pipelines, ETL/data workflows, and exploratory analysis using Python, TensorFlow, and Scikit-learn, with growing hands-on exposure to LLM and RAG-based applications using LangChain and Hugging Face.

EDUCATION

Master of Computer Applications (MCA) | Grade: 9.46/10 CGPA

Hindustan Institute of Technology & Science, Chennai

Aug 2024 – May 2026

Bachelor of Computer Applications (BCA) | Grade: 79.62% (7.69/10 CGPA)

Dr. M.G.R Chockalingam Arts College, Arni

Aug 2021 – May 2024

ACHIEVEMENTS

- ✓ Academic Rank Holder - MCA, Hindustan Institute of Technology & Science, Chennai (CGPA: 9.42 / 10).
- ✓ IEEE Conference Presenter - ICCRTEE 2026, Kalasalingam Academy of Research and Education, Tamil Nadu.
- ✓ IEEE Published Researcher - Brain Tumor Segmentation paper under IEEE Madras Section & IEEE Robotics and Automation Society.

PROJECTS & RESEARCH

Brain Tumor Segmentation in MRI Scans | Deep U-Net (IEEE Published)

ICCRTEE 2026 – IEEE Madras Section & IEEE Robotics and Automation Society

Feb - May 2026

- ✓ Designed and trained a U-Net deep learning model on the Kaggle LGG MRI dataset (3,000+ real-world medical samples).
- ✓ Built a custom OpenCV preprocessing pipeline with data augmentation; optimized using Dice Loss + Adam optimizer.

X-Ray Image Classification | MobileNet Transfer Learning (Medical AI)

Computer Vision – Healthcare Domain

Aug - Oct 2025

- ✓ Developed medical image classifier using MobileNet transfer learning with MD.ai DICOM API for real-world X-ray data integration.
- ✓ Achieved 85–90% validation accuracy using early stopping, learning rate scheduling, and fine-tuned hyperparameters.

Image Caption Generator | VGG16 + LSTM (Computer Vision + NLP)

Deep Learning – Multimodal AI

Jan - Apr 2025

- ✓ Built end-to-end multimodal pipeline combining VGG16 CNN image embeddings with LSTM-based sequence prediction.
- ✓ Trained on ~8,000 real-world image-caption pairs (MSCOCO); achieved BLEU score ~0.4 using a custom Keras data generator.

INTERNSHIPS & EXPERIENCE

Research Intern – Artificial Intelligence

Aug – Oct 2025

Hindustan Institute of Technology & Science, Chennai

- ✓ Built a Chest X-ray classification model using MobileNet, TensorFlow, and Grad-CAM for medical image analysis.

Data Science Intern

July 2025

CodeAlpha (Remote)

- ✓ Developed ML mini-projects using Python, Pandas, and Scikit-learn involving prediction, classification, and data analysis tasks.

KEY SKILLS

Languages & Tools	: Python, SQL, Git/GitHub, Jupyter Notebook, Google Colab
ML / DL Frameworks	: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib
LLM & GenAI (Learning)	: LangChain, LlamaIndex, OpenAI API, Hugging Face Transformers, Prompt Engineering, RAG (Retrieval-Augmented Generation)
Core ML/DL Expertise	: CNNs, LSTMs, U-Net, Transfer Learning, Image Segmentation, NLP, Feature Engineering
Data Engineering	: ETL Pipelines, Data Ingestion, Schema Modelling, Data Wrangling, MySQL Workbench
Data Analysis & BI	: Exploratory Data Analysis, Statistical Analysis, MS Excel, Power BI, Data Visualization (Matplotlib)
DSA Fundamentals	: Arrays, Linked Lists, Stacks, Queues, Trees, Hashing, Sorting & Searching
Web & Others	: HTML, CSS, JavaScript, MySQL
Professional	: Research Documentation, Analytical Thinking, Leadership, Communication

CERTIFICATIONS AND WORKSHOPS

Introduction to Algorithms and Analysis – Elite Certificate

NPTEL / IIT Kharagpur (SWAYAM)

Intellithon'25 – 24-Hour National Hackathon

Hindustan Institute of Technology & Science (Dept. of Intelligent Systems & Cybersecurity)

Cloud Partner: Zoho • 24-hour national-level event

Workshop: Practical Approaches to Data Analytics Using SQL and Cloud Services

Dept. of Computer Applications, Hindustan Institute of Technology & Science

LANGUAGES

Tamil (Native) | English (Professional)